

Disharee Bhowmick

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Education

Auburn University, Alabama

Ph.D. in Computer Science

Jan 2024 – present

GPA: 3.83

Auburn University, Alabama

Master of Science in Computer Science

Aug 2023 – May 2025

GPA: 3.83

University of Engineering and Management, Kolkata

Bachelor of Technology in Computer Science

Aug 2015 – Jul 2019

GPA: 3.6

Teaching Experience (GTA, Auburn University)

- Lab Instructor, Introduction to Engineering (ENGR 1110)
- Lead Teaching Assistant to Dr. Vishalini Ramnath, Introduction to Algorithms (COMP 3270)
- Teaching Assistant to Dr. Vishalini Ramnath, Formal Languages (COMP 4200)

Research Experience (GRA, Auburn University)

Graduate Research Assistant

CSSE Department, Auburn University | Advisor: Dr. Sathyanarayanan N. Aakur

Summer 2024 – present

Summer 2024 & Summer 2025 — Time Series Summarization (STaTS)

- Developed STaTS, a lightweight unsupervised framework for structure-aware temporal sequence summarization, adaptively compressing time series using a BIC-based multi-scale change detection criterion.
- Achieved up to 30x input compression while retaining 85-90% of full-resolution model accuracy across 150+ datasets spanning UCR, UEA, and ETT benchmarks.
- Work published in **Pattern Recognition Letters**, Vol. 202, pp. 135-140, 2026.

Summer 2026 — Adapting VLMs for Open World Event Segmentation

- Developing and evaluating temporal segmentation pipelines that adapt Vision-Language Models (VLMs) for open-world event segmentation, enabling recognition and localization of action categories beyond closed-set training distributions.

Publications

- Disharee Bhowmick, Ranjith Ramanathan, Sathyanarayanan N. Aakur. STaTS: Structure-aware temporal sequence summarization via statistical window merging. *Pattern Recognition Letters*, Vol. 202, pp. 135-140, 2026.
- Revanth Reddy Palem, Mariana E. Garcia-Ascolani, Disharee Bhowmick, David Lalman, Ranjith Ramanathan, Sathyanarayanan N. Aakur. Vision-Based Livestock Respiratory Monitoring for Smart Agriculture Sensing Systems. *IEEE Midwest Symposium on Circuits and Systems (MWSCAS)*, 2026.

Latest Preprints

- Shanmukha Vellamcheti, Uday Kiran Kothapalli, Disharee Bhowmick, Sathyanarayanan N. Aakur. CVT-Bench: Counterfactual Viewpoint Transformations Reveal Unstable Spatial Representations in Multimodal LLMs. *arXiv:2603.21114*, 2026.
- Revanth Reddy Palem, Disharee Bhowmick, Mariana E. Garcia-Ascolani, David L. Lalman, Ranjith Ramanathan, Sathyanarayanan N. Aakur. Physiological Signal-Aware Livestock Respiration Estimation from Video.

Employment History

Student Assistant — Innovation and Research Commons (I&RC)

Ralph Brown Draughon Library, Auburn University

Summer 2025

- Assisted patrons with multimedia, MakerSpace, and Adobe Creative Cloud resources, including 3D printing, VR/AR equipment, and digital media production tools.

Technician — Jay and Susie Gogue Performing Arts Center

Fall 2023

Auburn University

- Assisted with technical operations supporting live performances and events at the venue.

Cognizant Technology Solutions

Sep 2019 – Jun 2023

Programmer Analyst Trainee → Programmer Analyst → Associate | Pune, Maharashtra, India

- Completed Cognizant's graduate training in data analysis, statistics, and Python (Sep 2019 – Feb 2020).

Data Analytics (Mar 2020 – Feb 2021)

- Reduced recurring IT incidents by **25%** for enterprise clients by analyzing incident data in **SQL** and Python to pinpoint the few recurring root causes behind most ticket volume, and working with operations teams to prioritize permanent fixes.
- Tracked the impact of those fixes, comparing monthly incident counts before and after, by building Power BI and Tableau dashboards reviewed with technical and business stakeholders.

Machine Learning, Internal Project (Mar 2021 – Feb 2022)

- Led development of a predictive analytics model on 100K+ records, selecting the final approach by benchmarking supervised and ensemble models in Python (scikit-learn, pandas).
- Performed feature engineering, data cleaning, and model validation across multiple ML architectures.

Generative AI (Mar 2022 – Jun 2023)

- Deployed LLM solutions to production in client workflows, integrated with downstream scoring logic, by serving open-source LLMs on **AWS SageMaker** and GPT models on **Azure OpenAI**, packaged with Docker, Git, and CI/CD.
- Improved answer accuracy and reduced hallucinations by building **Retrieval-Augmented Generation (RAG)** pipelines over enterprise knowledge bases (document parsing, chunking, embeddings, hybrid keyword + vector search), validated through RAG evaluation of retrieval relevance and answer grounding.
- Adapted pretrained LLMs to domain-specific tasks by **fine-tuning** them on curated domain datasets.
- Met enterprise compliance standards by adding OCR-based document parsing with PII redaction, along with model monitoring, logging, and metadata/lineage tracking.

Intern — Ogma Tech Lab

Jul 2018 – Aug 2018

Kolkata, India

- Developed an Android Studio-based "Puja Parikrama" application using Android Studio and Firebase.

Skills & Courses

- Mathematics & Statistics: Basic statistics, probability theory, statistical tests, time series concepts (autocorrelation, seasonality, stationarity)
- Programming: Python, NumPy, pandas, scikit-learn, TensorFlow, PyTorch
- Data & ML: Cleaning, missing data handling, feature engineering, supervised/unsupervised learning, model evaluation, ensembles
- ML Frameworks: RNNs, LSTMs, GRUs, CNNs, Attention, Transformers
- Data Visualization: Power BI, Tableau
- GenAI & CV: Large Language Models, vision-based generative AI, computer vision pipelines
- Courses: Deep Learning, Computational Imaging, Machine Learning, Image & Video Processing, Natural Language Processing, Multimedia & Web Databases

Honors and Awards

- Recipient of the 100+ Women Strong Outstanding Departmental Annual Graduate Award (2026), Samuel Ginn College of Engineering, Auburn University — selected as one of two graduate student honorees from the Department of Computer Science and Software Engineering. Award value: \$2,000.
- Only CSSE nominee for the ACM SIGHPC Fellowship by Auburn University.
- Received Rising Star Award for project work from Cognizant Technology Solutions.
- Awarded Star of the Month multiple times at Cognizant.